

Francesco Dorati

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Research interests: Physical AI · Computer Vision · Reinforcement Learning · Autonomous Systems

Education

Politecnico di Milano – <i>MSc in Computer Science and Engineering</i> Specialization in Artificial Intelligence	Milan, Italy 2025 - Present
Beijing Institute of Technology – <i>Summer Program</i> Intelligent Robotics Engineering	Zhuhai, China Jul 2026
UPC Barcelona – <i>UNITECH Exchange</i> Courses: <i>Computer Vision, Machine Learning, NLP, Advanced Processor Architecture</i>	Barcelona, Spain Feb – Jun 2026
Politecnico di Milano – <i>BSc in Computing Systems Engineering</i> Final Grade: 100/110 Final Projects (Avg. Score: 98.6%): <i>Algorithms & Data Structures (C), Software Engineering (Java), FPGA Design (VHDL)</i>	Milan, Italy 2022 - 2025

Publications

Foundation ViTs Can Learn Where to Act: Task Affordance Segmentation and Surface Normals

Francesco Dorati, Tommaso F. Banfi
ECCV 2026 Workshop OpenSUN3D – **top 15%** of accepted papers

Bayesian Optimization for Learning Nonlinear MPC in Autonomous Agent Navigation

Lorenzo Ortolani, Gabriel Voss, Gabriele Beltrami, **Francesco Dorati**, Tommaso F. Banfi
IEEE ICRA 2026 Workshop

Optimizing Initial Path Finding in Informed-RRT* with a Novel Map-Adaptive Sampling Technique

Tommaso F. Banfi, **Francesco Dorati**, Nicola Manzoni, Jesús Martínez-Gómez
7th Iberian Robotics Conference (ROBOT 2024), IEEE Xplore

Experience & Research

AI Researcher <i>Talos Robotics AI</i> — <i>Early-stage humanoid robotics startup</i> Contributing to a platform for training, data collection, and deployment of VLA manipulation policies on humanoid robots for industrial environments; team secured 25k industrial research contract (VHIT) and 10k award (TEF Ignition Program). - Developing a multi-task perception pipeline for zero-shot robotic affordance detection: predicts 7-class pixel-level affordance masks and surface normals from RGB using DINOv2 with multi-scale feature fusion and RGB skip connections 80.3% mean-IoU on the official UMD benchmark, surpassing prior SOTA (Mask2Former, 75.9%)	Oct 2025 – Present Milan, Italy
R&D Intern: Hierarchical Navigation Stack for Industrial AGV <i>DevNut (Italian tech startup)</i> Sensor-based AGV navigation in Webots , replacing fixed floor-line infrastructure - Designed a two-layer planner: global visibility-graph + A* for path planning using local occupancy-grid planner from LiDAR - Designed a map-differencing approach separating static walls from dynamic obstacles, with per-frame cluster extraction and velocity-aware tracking	Jan 2026 – Jul 2026 Remote
Undergraduate Researcher – FALCO Autonomous Drone <i>Automation Engineering Association, Politecnico di Milano</i> Developed novel path planning algorithm for autonomous drone navigation in 3D environments - Achieved 85% faster initial path discovery and 78% reduction in node count than state-of-the-art Informed-RRT* - Proposed a map-adaptive probabilistic sampling strategy and implemented the full algorithm in C++, with a Python benchmarking and visualization framework	Nov 2023 – Feb 2025 Milan, Italy

Awards & Fellowships

UNITECH International Programme – cohort 2025 Competitive international programme combining academic exchange and industry internship	2025 – 2026
Tiny_Hack Competition – 3rd place Embedded computer vision on microcontrollers	2025
Best Freshmen Award – Politecnico di Milano	2023

Technical Skills

- **Programming:** Python, C++, C, Java, MATLAB, SQL
- **ML & Deep Learning:** PyTorch, Scikit-learn, Model Quantization, DINOv2
- **Computer Vision:** OpenCV, YOLO, Image Segmentation, Object Tracking, Sensor Fusion
- **Robotics & RL:** ROS/ROS2, PPO, Actor-Critic, Imitation Learning, Isaac Sim, MuJoCo, Gazebo
- **Hardware & Embedded:** VHDL/FPGA, Raspberry Pi, Arduino
- **Infrastructure:** Linux, Docker, Git, Edge Deployment

Courses & Certifications

Machine Learning Specialization – Stanford University, DeepLearning.AI [link]	2025
Deep Learning Specialization – DeepLearning.AI [link]	2024
CS50x: Introduction to Computer Science – Harvard University [link]	2021

Languages

English – IELTS Academic 7.5; Cambridge IGCSE grade A	Fluent
Italian	Native